

# Problem 10

**Problem 1** For the piecewise function  $f$  defined by

$$f(x) = \begin{cases} \sin(x) & \text{if } x \leq 0 \\ \frac{x^2}{x^2 - 4} & \text{if } 0 < x < 2 \\ \frac{x^2}{x^2 + 4} & \text{if } 2 \leq x \end{cases}$$

- (a) Find all vertical asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.

**Explanation.** Candidates for vertical asymptote:  $x^2 - 4 = 0 \implies x = -2, x = 2$  (Each piece of  $f$  is continuous at all other points).

Note:  $x = -2$  is not a vertical asymptote, since  $f(x) = \sin(x)$  near  $-2$ , so  $\lim_{x \rightarrow -2} f(x) = \lim_{x \rightarrow -2} \sin(x) = \sin(-2)$ .

$$\lim_{x \rightarrow 2^-} \frac{x^2}{x^2 - 4} = \lim_{x \rightarrow 2^-} \frac{x^2}{(x - 2)(x + 2)} = -\infty$$

Note: the limit is of the form  $\frac{\#}{0}$ , the numerator positive, and the denominator negative.

The line  $x = 2$  is a vertical asymptote of  $f$  because  $\lim_{x \rightarrow 2^-} f(x) = -\infty$ . This is the only vertical asymptote of  $f$ .

- (b) Find all horizontal asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.

**Explanation.** Check End Behavior as  $x \rightarrow \infty$ :

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{x^2}{x^2 + 4} \text{ is of the form: } \frac{\infty}{\infty} \\ \lim_{x \rightarrow \infty} \frac{x^2}{x^2 + 4} = \lim_{x \rightarrow \infty} \frac{x^2}{x^2 + 4} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}} \\ = \lim_{x \rightarrow \infty} \frac{1}{1 + \frac{4}{x^2}} \\ = 1 \end{aligned}$$

*Problem 10*

The line  $y = 1$  is a horizontal asymptote of  $f$  since  $\lim_{x \rightarrow \infty} f(x) = 1$ .

Check End Behavior as  $x \rightarrow -\infty$ :

$\lim_{x \rightarrow -\infty} \sin(x)$  does not exist, since the function oscillates.

For example,  $f(-n\pi) = \sin(-n\pi) = 0$ , while  $f(-(2n+1)\frac{\pi}{2}) = \sin\left(-(2n+1)\frac{\pi}{2}\right) = 1$ , for all positive integers  $n$ .

There is no horizontal asymptote as  $x \rightarrow -\infty$  since  $\lim_{x \rightarrow -\infty} f(x)$  does not exist.